

Artefact 1 Report: Data Ingestion Pipeline

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CAPSTONE AND TEAM INTEGRATION STATEMENT

Our team is developing a hybrid AI system to predict periods of electricity-network stress in NSW using demand, renewable generation, weather, and grid data. My contribution focuses on exploratory data analysis, data cleaning, and ingestion, providing quality control and supporting downstream feature engineering and model development.

PROJECT PROPOSAL

PROBLEM STATEMENT

NSW's electricity system faces a growing challenge as electricity demand increases while the state's generation mix undergoes significant transformation. Transgrid, the operator of the electricity network in the state, forecasts that data centres alone could increase from approximately 3% of NSW peak electricity demand today to around 20% by 2035. At the same time, more than 80% of the state's coal-fired generation capacity is expected to retire over a similar period, while the electricity system continues its rapid transition towards wind and solar generation [1].

These changes increase the importance of predicting periods of energy shortfall and network stress to maintain system reliability and security [2]. However, prediction alone is insufficient; operators also require information about contributing factors and appropriate mitigation strategies. This research therefore investigates whether a hybrid AI system can integrate demand, renewable generation, weather, and grid-condition data to predict network stress and recommend appropriate responses.

PROJECT OBJECTIVES

The intended objectives of the AI system are as follows:

- Analyse electricity demand, renewable energy generation, weather, and grid-condition data as key input features.
- Ingest, clean, transform, and integrate these input features through an ETL pipeline.
- Accurately forecast electricity demand and identify potential periods of network stress using machine learning algorithms.
- Apply logical inference to interpret model outputs

JUSTIFICATION

This problem is well suited to a hybrid AI system because machine learning can identify complex, non-linear patterns in time-series data for forecasting, while logical inference can apply expert knowledge and operational constraints. Combining both approaches can reduce physically implausible predictions, improve robustness, and support more reliable recommendations [3].

MODEL SURVEY

Several hybrid approaches were considered for predicting electricity-network stress.

Table 1: Analysis of different hybrid systems

Approach	Description	Advantages	Limitations
LSTM + rule-based expert system	Combines pattern-recognition and temporal-processing power of Long Short-Term Memory (LSTM) with a rule-based expert system which evaluates forecast.	Strong at modelling temporal and non-linear relationships in electricity demand; expert rules provide transparent domain reasoning and can improve forecasting/decision support [4].	Requires substantial historical training data; rule development depends on domain experts and rule bases can become difficult to maintain.
XGBoost + rule-based system	Learns relationships between input features and target value, rather than treating the data as a time sequence. Uses a separate logic rule engine to interpret the prediction.	High forecasting accuracy on tabular data, requires less training compared with LSTM, and faster/simpler to train [5].	Requires explicit temporal feature engineering such as rolling-window features and lag.
ANFIS	Model that merges neural networks and fuzzy logic. Fuzzy logic is an approach to data processing that allows for degrees of truth between 0 and 1, rather than strict binary values [6].	Can model complex nonlinear relationships and uncertain concepts such as “high demand” and “low renewable generation”.	As input features and membership functions increase, the possible fuzzy rules can grow exponentially.

Based on the analysis in Table 1, the LSTM + rule-based expert system approach is proposed. This architecture was selected since our problem statement is a valid time-series problem, making it perfect for LSTM’s temporal modelling.

HUMAN FACTORS

A human-in-the-loop approach will be adopted, where the system predicts periods of network stress and provides recommendations, while the end user remains responsible for deciding how to act on this information. The system should also include a level of explainability, allowing users to understand the key factors and reasoning behind each prediction and recommendation.

ETHICS REVIEW

The main ethical considerations for the system are safety and reliability risks arising from inaccurate AI predictions and recommendations. To address these risks, appropriate model validation, conservative decision thresholds, and extensive testing will be used during the early stages of development to reduce the likelihood of errors propagating through the system.

STAKEHOLDER ANALYSIS AND CONFLICTING REQUIREMENTS

The two primary stakeholders for this system are energy grid operators and large energy consumers.

Table 2: Stakeholder analysis

Stakeholder	Interest	Power
Grid Operator	Require timely information about expected periods of grid stress and appropriate actions that could be taken in response.	High power since they directly manage grid operations and make operational decisions based on system conditions.
Energy Consumers	Want to understand changes in the NSW electricity system and the reasons behind actions that may affect energy availability, reliability, or demand-management requirements.	Low power since they do not directly control the grid but may provide valuable insight into the downstream impacts of operational decisions

A key consideration is balancing operators' decision-support requirements with consumers' need for transparency. The system should therefore retain historical predictions, recommendations, and supporting explanations for appropriate stakeholder review.

DATASET ANALYSIS

SOURCE

The Open Electricity dataset was selected for this project since it is open-source, tracks energy demand, and contains a wealth of historical data for Australia's grid since 1999. Open Electricity supports various time intervals such as 5 – minutes to 1 hour which makes it particularly useful for time series analysis. Additionally, there is an official Open Electricity Python SDK which makes it convenient to access the API and fetch data during ingestion.

Platform: <https://platform.openelectricity.org.au/>

Documentation: <https://docs.openelectricity.org.au/introduction/>

EXPLANATORY DATA ANALYSIS (EDA)

The dataset contains 234,312 rows and 27 columns of hourly electricity market and network data from 1 January 2000 to 24 September 2026. Initial validation confirmed appropriate data types.

One of the main challenges during data cleaning was determining whether unusual or missing values represented invalid data or reflected legitimate characteristics of the dataset. This required additional domain knowledge and investigation. For example, as shown in Figure 1, the `storage_battery` column contained approximately 96% missing values. Further investigation found that collection of this metric only began after 30 June 2025 (see Appendix 1). Other columns also contained missing values; however, these generally represented periods where data was unavailable rather than invalid observations. Because preserving the temporal structure of the dataset was important, many of these missing values were retained rather than removing entire rows.

	Missing count	Missing (%)
<code>storage_battery</code>	225080	96.0600
<code>flow_exports</code>	83271	35.5385
<code>flow_imports_energy</code>	83271	35.5385
<code>flow_exports_energy</code>	83271	35.5385
<code>flow_imports</code>	83271	35.5385
<code>demand_gross_energy</code>	56567	24.1417
<code>demand_energy</code>	46038	19.6482
<code>renewable_proportion</code>	10532	4.4949
<code>demand_gross</code>	10532	4.4949
<code>renewable_with_storage_proportion</code>	10532	4.4949
<code>generation_renewable_energy</code>	452	0.1929
<code>generation_renewable</code>	452	0.1929
<code>generation_renewable_with_storage</code>	452	0.1929
<code>generation_renewable_with_storage_energy</code>	452	0.1929
<code>demand</code>	10	0.0043
<code>power</code>	9	0.0038
<code>market_value</code>	9	0.0038
<code>energy</code>	9	0.0038
<code>emissions</code>	9	0.0038

Figure 1: Missing value counts and percentages

EDA also identified an issue affecting renewable energy and other market-related energy metrics in the year 2000, where values appeared to have been incorrectly scaled by a factor of 1,000. Although extensive investigation did not identify the exact root cause, Open Electricity had previously acknowledged the issue. This required additional consideration during data cleaning to ensure that these anomalous values did not negatively affect subsequent analysis or model training. The distribution of negative curtailment values was also examined (see Appendix 3). Overall, EDA revealed that the dataset was relatively clean, with only minimal preprocessing required.

SYSTEM ARCHITECTURE AND IMPLEMENTATION

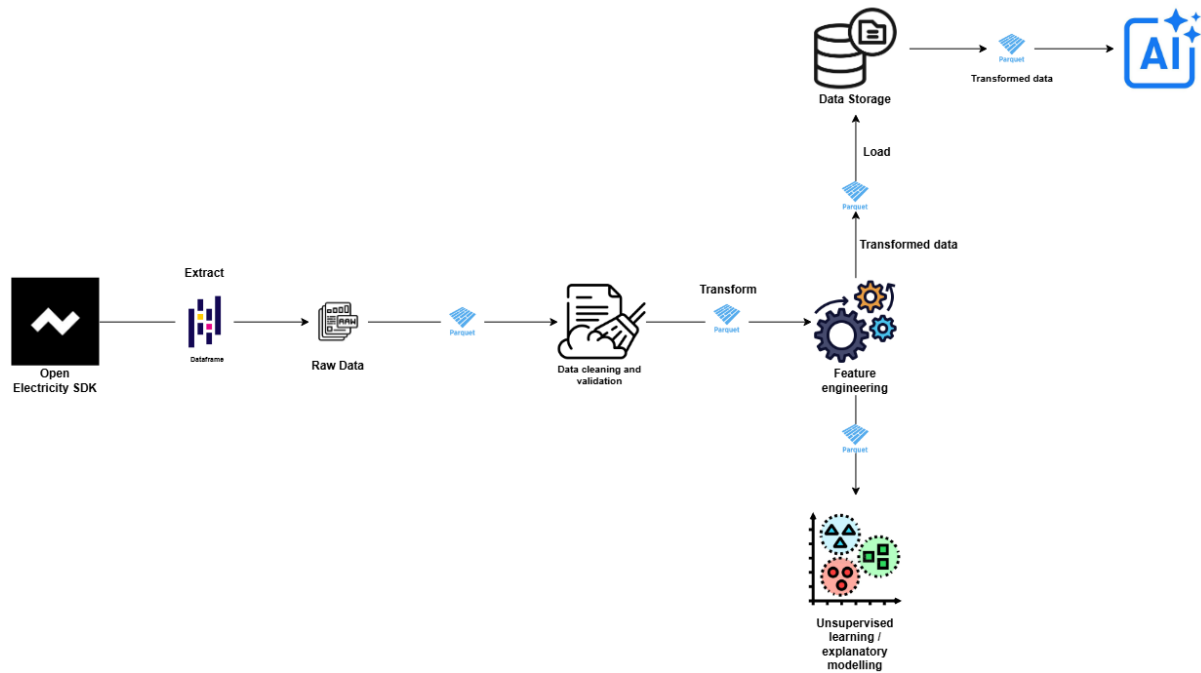


Figure 2: Data ingestion pipeline

Figure 2 illustrates the system’s automated data ingestion pipeline. To protect API credentials, the pipeline uses environment-based configuration and HTTPS bearer-token authentication. Data integrity is maintained through schema, data-type, timestamp, duplicate-record, and temporal-continuity validation. If any integrity check fails, the pipeline raises an exception and terminates to prevent invalid or potentially corrupted data from progressing to subsequent processing stages.

PROCESS

The pipeline automatically retrieves hourly market and network data for NSW from the OpenElectricity API in 30-day chunks. It removes overlapping records, restructures the responses, merges market and network data by timestamp, and saves the raw dataset locally in Parquet format. Parquet was selected because it efficiently stores large volumes of numerical time-series data while preserving data types.

During cleaning, the pipeline confirms that the dataset is not empty and validates its required fields, numeric types, timestamps, duplicate records, infinite values, and hourly continuity. Documented corrections are then applied, while missing and unusual values are retained for investigation rather than replaced without justification.

Preprocessing restricts the dataset to the period with valid curtailment coverage, removes redundant fields, interpolates gaps of up to three hours, and flags imputed and negative-curtailment values. The pipeline then creates cyclical time, lag, and rolling-statistic features. Selected variables are standardised and reduced using PCA before clustering and anomaly detection (see Appendix 2 and 4).

Each stage is saved separately during development to support reproducibility, debugging, and auditing. In production, storage could be reduced by retaining only the immutable raw dataset, final transformed dataset, and associated audit reports; temporary intermediate datasets could be discarded after successful validation.

TECHNICAL DECISIONS

Pandas and NumPy are no-brainers to use since they are perfect for efficient ingestion and vectorized processing. Data is batch processed in 30-day chunks to control memory use and comply with API limits.

Scikit-learn supports standardisation, PCA, clustering, and Isolation Forest anomaly detection. PCA reduces computational cost, while Isolation Forest identifies unusual behaviour without labelled threat data.

The current pipeline is designed to analyse near-real-time rather than immediate detection. In production, historical data would be used for batch model training. Once trained, the model could evaluate each new observation individually, which would be significantly faster than repeatedly processing the entire historical dataset. More discussion of this will be present in A2.

REFERENCES

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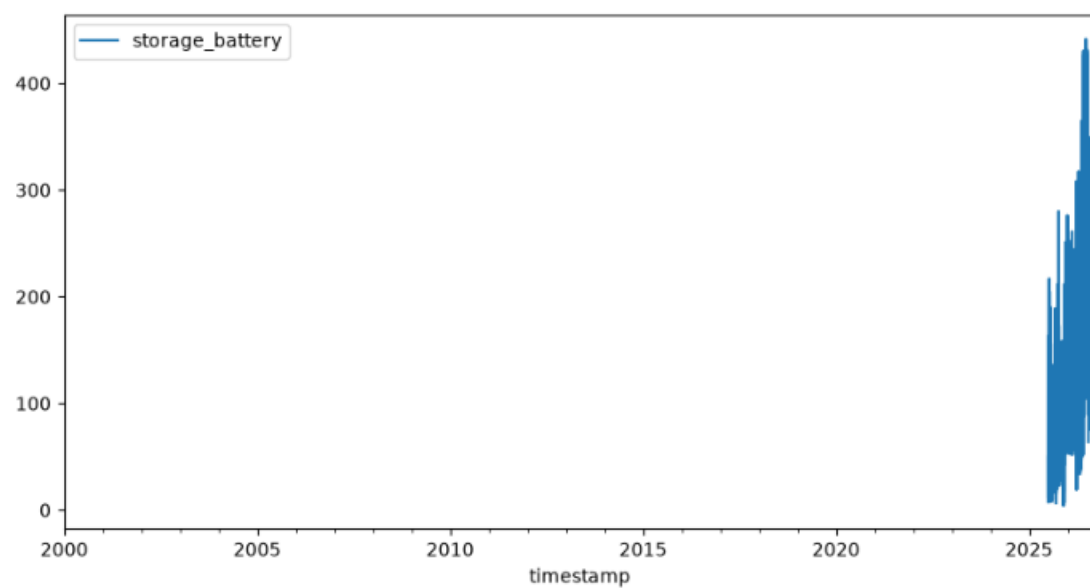
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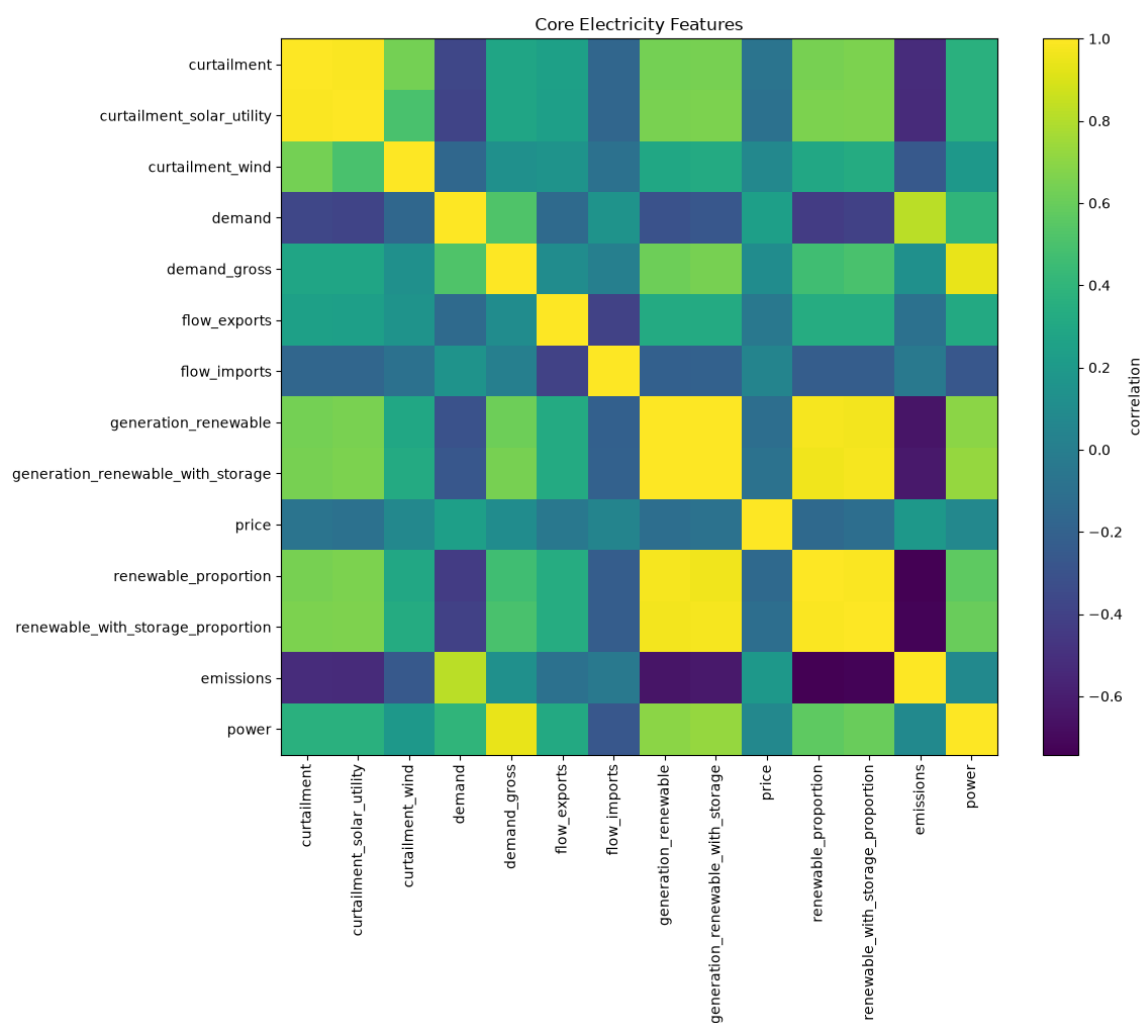
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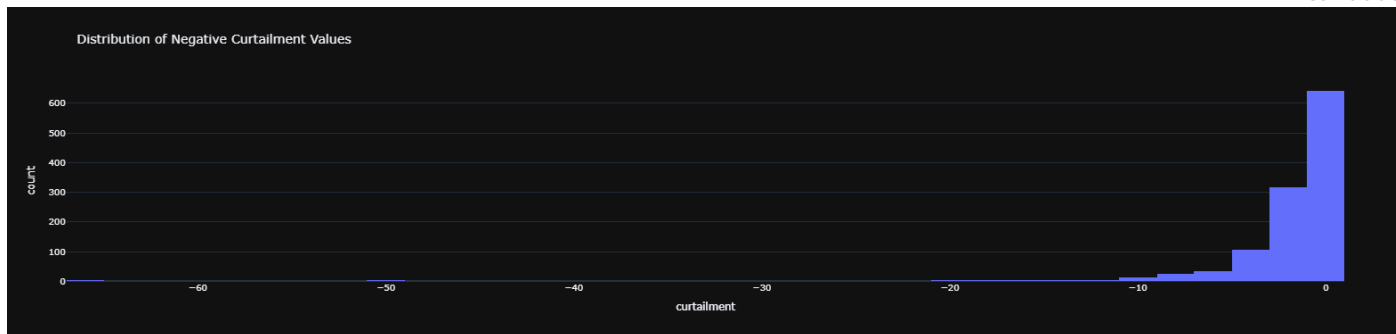
APPENDICES



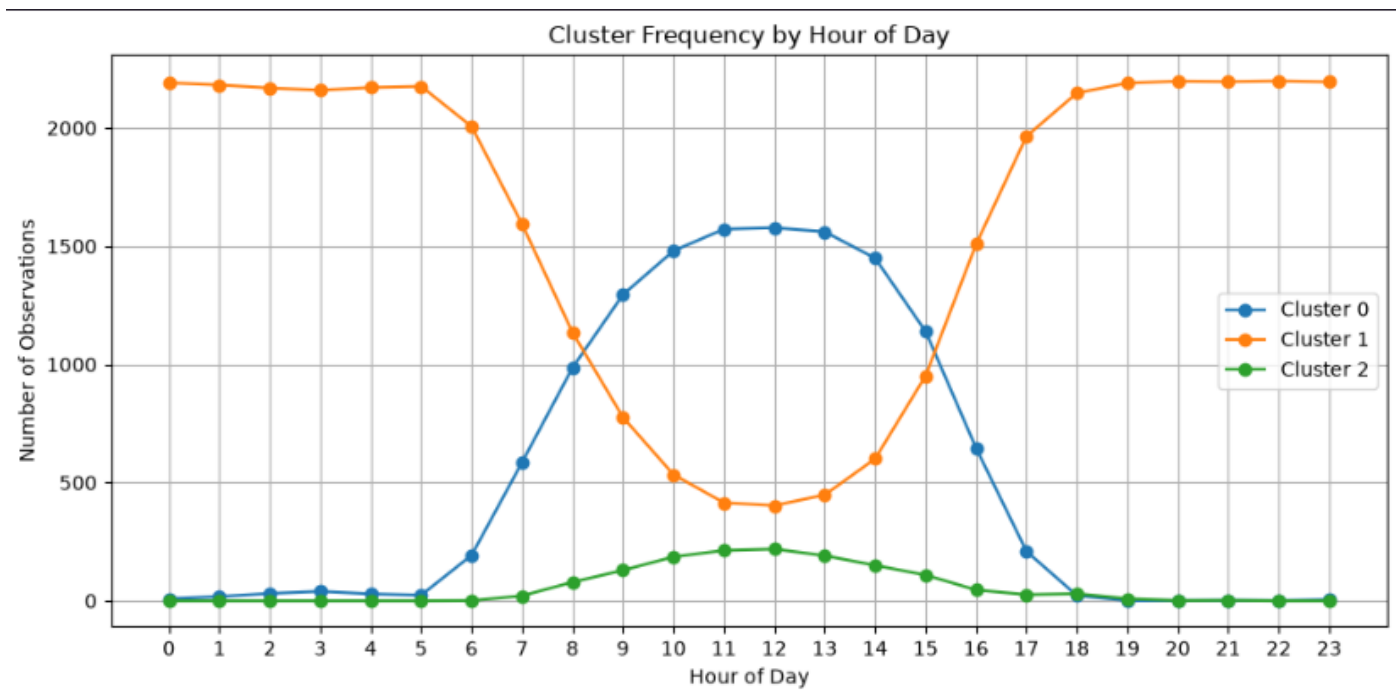
Appendix 1: Line plot of complete values in storage_battery



Appendix 2: Correlation matrix heatmap



Appendix 3: Distribution of negative curtailment values



Appendix 4: Cluster frequency by hour of day

APPENDIX: AI USAGE DECLARATION

I acknowledge the use of OpenAI GPT-5.6 Sol for research and some uses of code prompting in the assignment. ChatGPT was used for light paraphrasing and rewording in this report, however, all content is my own.

Full prompt outputs can be found in prompts.md under the “Misc” folder.

No.	Question	Summary of output
1	What steps should be followed to clean and perform exploratory analysis on the electricity dataset?	Proposed a reproducible workflow covering the modelling objective, raw-data preservation, data dictionaries, quality auditing, timestamp handling, missing data, range checks, exploratory analysis, leakage prevention, and reporting.

2	What are the sources, types, units, expected ranges, and meanings of missing values for the selected OpenElectricity metrics?	Produced a sourced data dictionary for all 26 metrics and clarified valid negative values, percentages above 100%, structural historical missingness, and why missing values must not automatically be replaced with zero.
3	Which storage format is most suitable for the multi-year hourly dataset?	Compared Parquet, CSV, compressed CSV, SQLite, Feather, Pickle, Excel, and JSON, then recommended Parquet for working datasets and CSV for human-readable summaries or compatibility exports.
4	How does the ingestion pipeline address cybersecurity, and what weaknesses remain?	Reviewed confidentiality, integrity, availability, auditability, credential handling, validation, resilience, file integrity, and dependency security, then recommended controls such as secret scanning, retries, unit validation, atomic writes, checksums, and vulnerability scanning.
5	Which hybrid AI architecture is most suitable for forecasting electricity-network stress and recommending responses?	Compared LSTM- and XGBoost-based expert systems, ANFIS, and deep neuro-symbolic models, then recommended an LSTM with a rule-based expert system for accurate forecasting and explainable recommendations.